GROWRIGHT AFRIQUE: AI-DRIVEN CROP & SOIL INTELLIGENCE SYSTEM

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HIGHLIGHTS

- Project Overview
- Business Understanding: Business Problem and Objectives
- Data Understanding
- Data Preparation
- Results
- Recommendations
- Limitations and Suggestions for further studies

PROJECT OVERVIEW

- Farmers struggle with crop yield prediction and fertilizer selection.
  - Driven by climate change, soil degradation, and rising input costs.
- Current stages of soil prediction involves: Collection soil sample:
  - Assessment at KALRO (Kenya Agricultural and Livestock Research Organization).
    - With long waiting times – 3 weeks.
  - Local tools exist BUT rely heavily on devices (Soilpal)
    - Limited accessibility for rural farmers with low resources.

PROJECT GAP

- Current tools depend on expensive chemical soil analysis.
  - Lack of user-friendly AI tools for farmers.
  - No strong image-based soil analysis systems widely available.
- We therefore:
  - Plan to leverage on the use of physical characteristics as well as soil images.
    - Easier for farmers to observe and don't need instruments.
  - By developing AI algorithms and hosting on easy to access Websites and phone apps.

BUSINESS QUESTIONS

- How effectively can crop type be predicted by combining soil images with field-observed physical soil attributes?
  1. How accurately can crop type be predicted using field-observed physical soil indicators (erosion signs, slope characteristics, surface stones, and crust formation)?
  2. How accurately can crop type be predicted using soil images?

STAKEHOLDERS WHO MAY BENEFIT

- ✓ Local farmers – to predict:
  - ✓ Type of crop based on:
    - ✓ Physical Soil Attributes: Erosion type, surface salt, surface crust, surface stone e.t.c
  - ✓ Type of fertilizer to augment with: inorganic, crop residue, green manure, animal.
- ✓ Ministry of agriculture – for fertilizer imports and location deployment.
- ✓ Agricultural Food Authority – for yield predictions based on location.

DATA UNDERSTANDING

- Dataset sourced from Soil4Africa (ISRIC platform) with two datasets:
  - 1st: Contains Physical dataset with 54 features + crop type target.
  - 2nd: Contains Image dataset with soil photographs.
    - Data merged using common plot_code identifier.
- Target Variable: Crop type.
- Feature Variables: 54 columns with features divided into: geographic, physical soil attributes, land management practices and grazing intensity indicators.
  - We focused solely on the physical soil attributes.

LIBRARIES IMPORTED AND USED

- We used several libraries:

1. Pandas, NumPy – Data cleaning & numerical processing.
2. Matplotlib, Seaborn – Data visualization & insights.
3. Base64, PIL, BytesIO – Image decoding & processing.
4. Scikit-learn – ML models, preprocessing & evaluation.
5. TensorFlow / Keras – Deep learning (CNN & Neural Networks).
6. Joblib, OS – Model saving & deployment.

DATA CLEANING – 1ST DATASET ON PHYSICAL ATTRIBUTES

- Started with 13876 entries with 55 columns:
- Removed columns with high missing values (>41%).
  - With drop from 55 columns to 39.
- Dropped redundant duplicate columns.
  - With drop from 39 columns to 37.
- Removed remaining null rows.
  - With drop of entries from 13876 to 2467.

DATA CLEANING – 2ND DATASET ON SOIL IMAGES

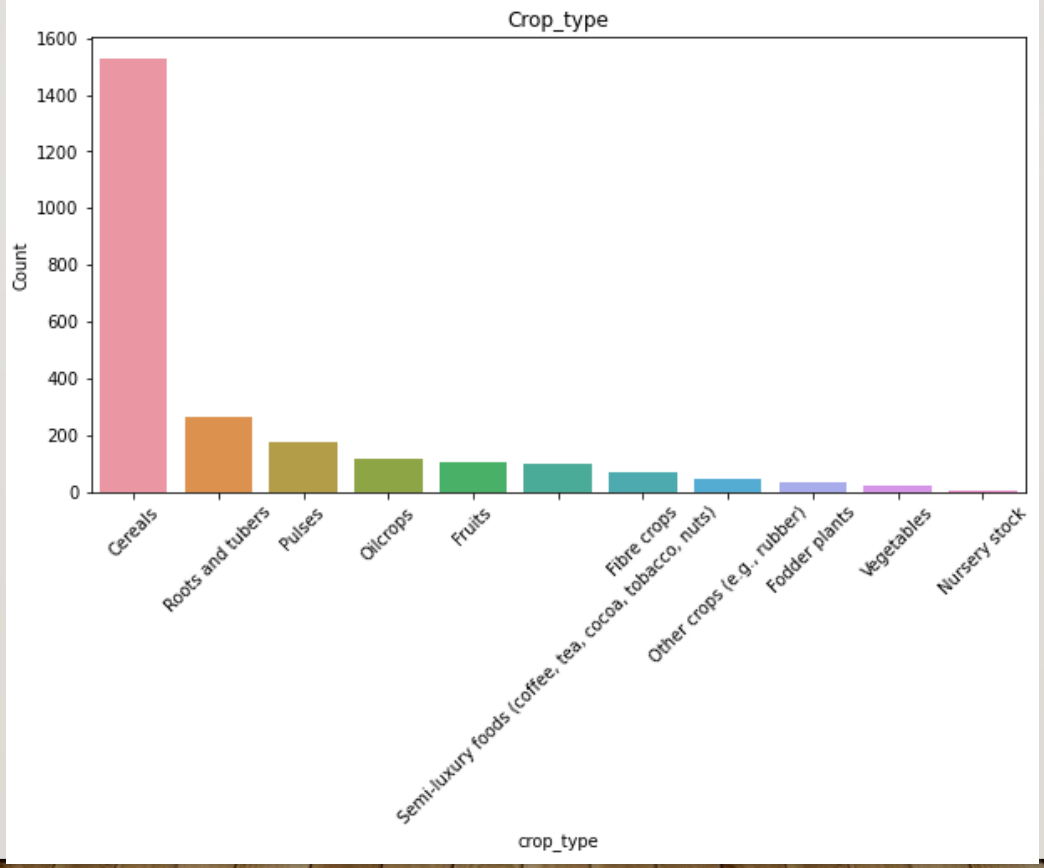
- Scraped soil images using Selenium automation.
- Confirmed accurate loading of the images in base 64 format.
  - Had horizontal_up, horizontal_down, vertical photos.
- Filtered images and kept only "horizontal_up".
- Decoded the images and converted them to numpy arrays for ML modelling.
- Dropped the horizontal_up base_64 columns.
- Confirmed null missing and duplicated values.

FINAL OUTPUT FROM DATA CLEANING

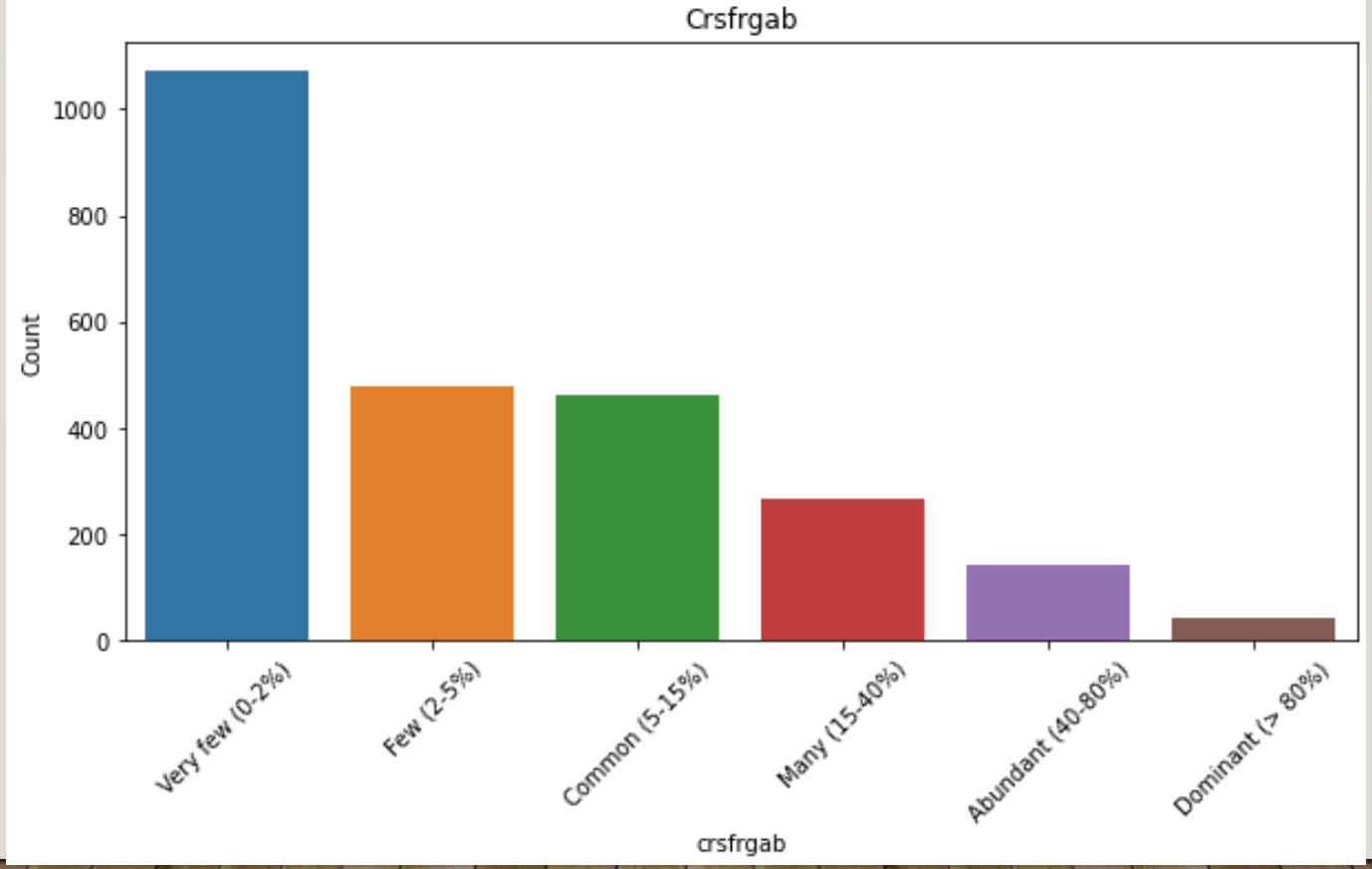
- From the 1st dataset on physical soil attributes:
  - 2467 entries with 37 columns (including identifier: plot_id)
- From the 2nd dataset on soil images:
  - 396 images with only numpy arrays of the horizontal_up photos and plot_id (identifier)
- Concatenated the 2 dataset to have 1 unified Kenyan dataset for image classification:
  - 87 entries with 38 columns.

RESULTS

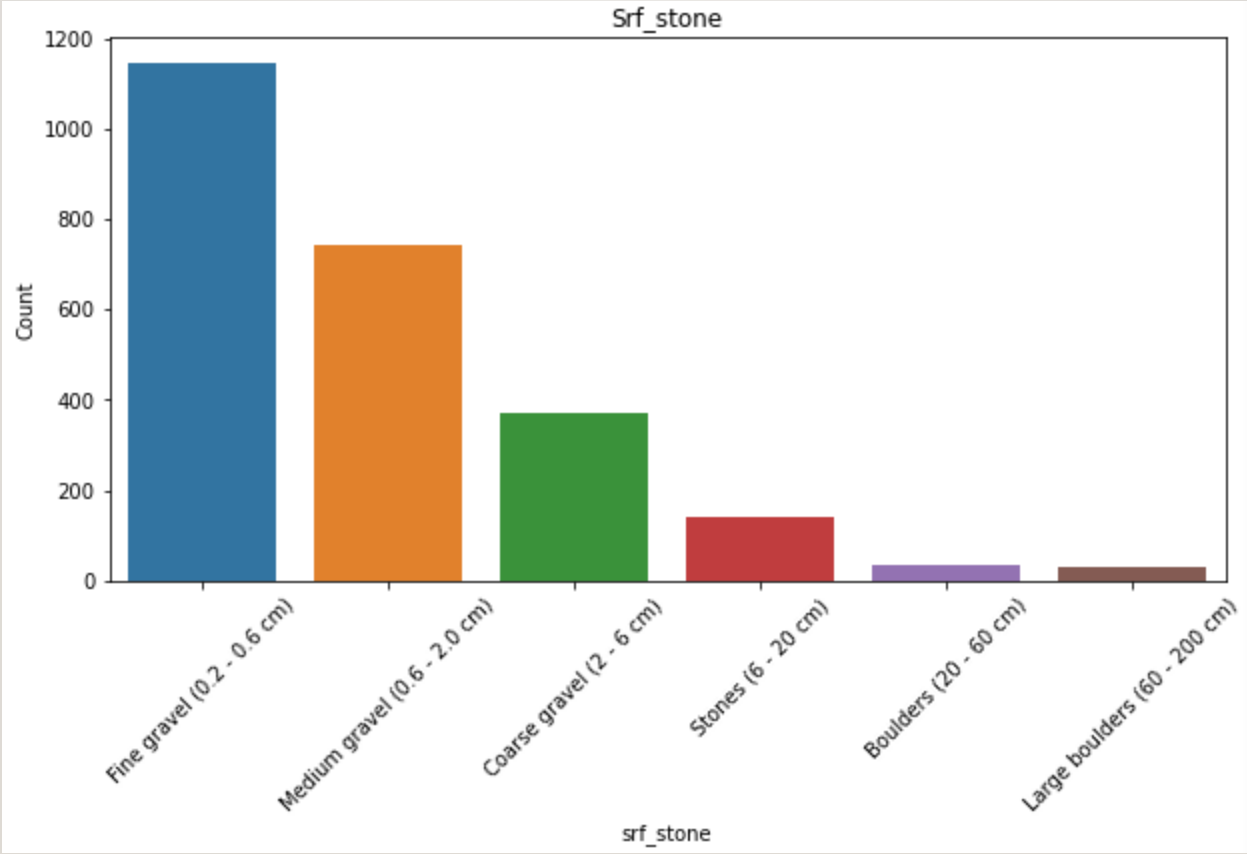
CROP TYPE


Crop_type
Count
1600
1400
1200
1000
800
600
400
200
0
Cereals
Roots and tubers
Pulses
Oilcrops
Fruits
Fibre crops
Semi-luxury foods (coffee, tea, cocoa, tobacco, nuts)
Other crops (e.g., rubber)
Fodder plants
Vegetables
Nursery stock
crop_type

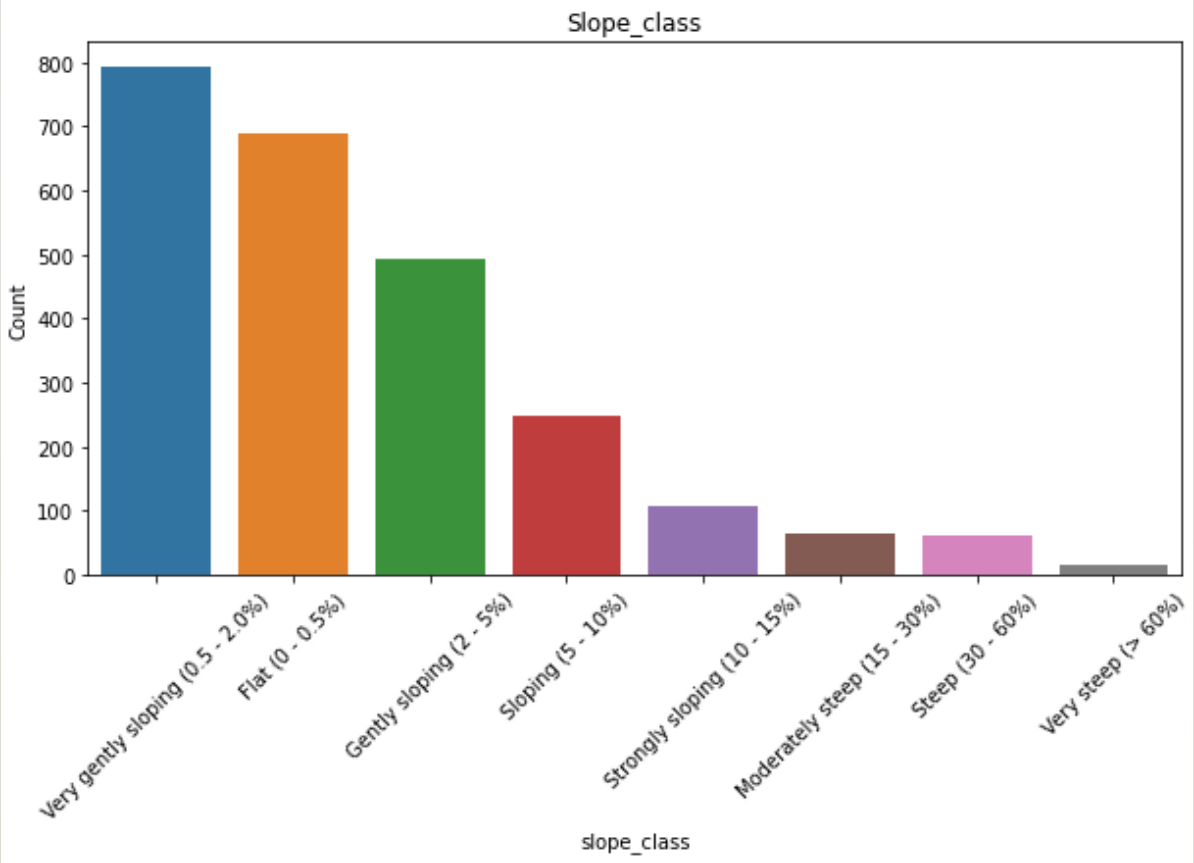
COARSE FRAGMENTS ABUNDANCE



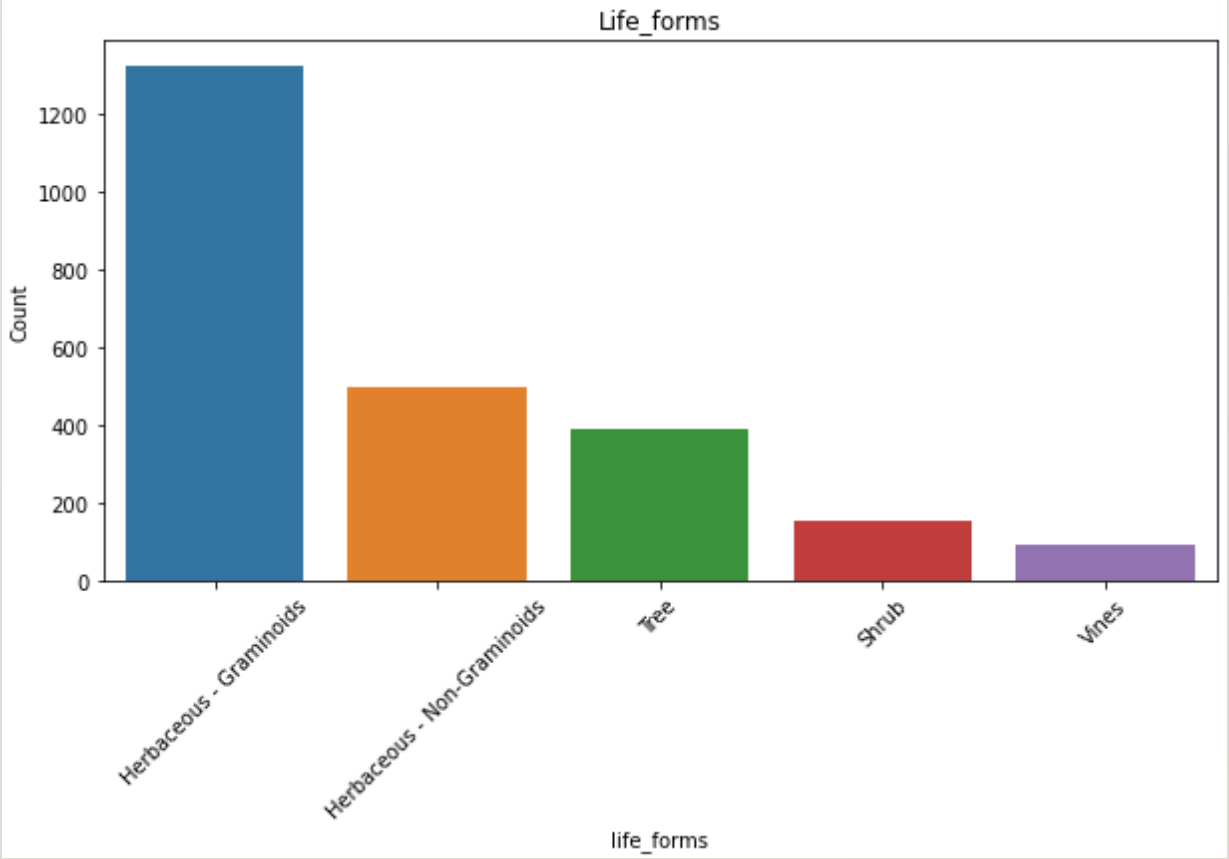
SURFACE STONE



SLOPE CLASS


Slope_class
Count
800
700
600
500
400
300
200
100
0
Very gently sloping (0.5 - 2.0%)
Flat (0 - 0.5%)
Gently sloping (2 - 5%)
Sloping (5 - 10%)
Strongly sloping (10 - 15%)
Moderately steep (15 - 30%)
Steep (30 - 60%)
Very steep (> 60%)
slope_class

LIFE FORMS


Life_forms
Count
1200
1000
800
600
400
200
0
Herbaceous - Graminoids
Herbaceous - Non-Graminoids
Tree
Shrub
Vines
life_forms

MOST LAND

- Had:
  - No signs of erosion.
  - No surface crust.
  - No surface stone
  - No Bare soil.

- Was level land and sloping land. Very few was steep land.

DETERMINING WHICH PHYSICAL ATTRIBUTES PREDICTED CROP TYPE BETTER.

- **Vegetation factors dominate crop prediction accuracy.**
  - Herbaceous (Non-Graminoids) – 0.79 (strongest predictor)
  - Tree-based vegetation – 0.73
  - Vines presence – 0.71
  - Shrub vegetation – 0.65
- Flooded / aquatic cultivated land – 0.50

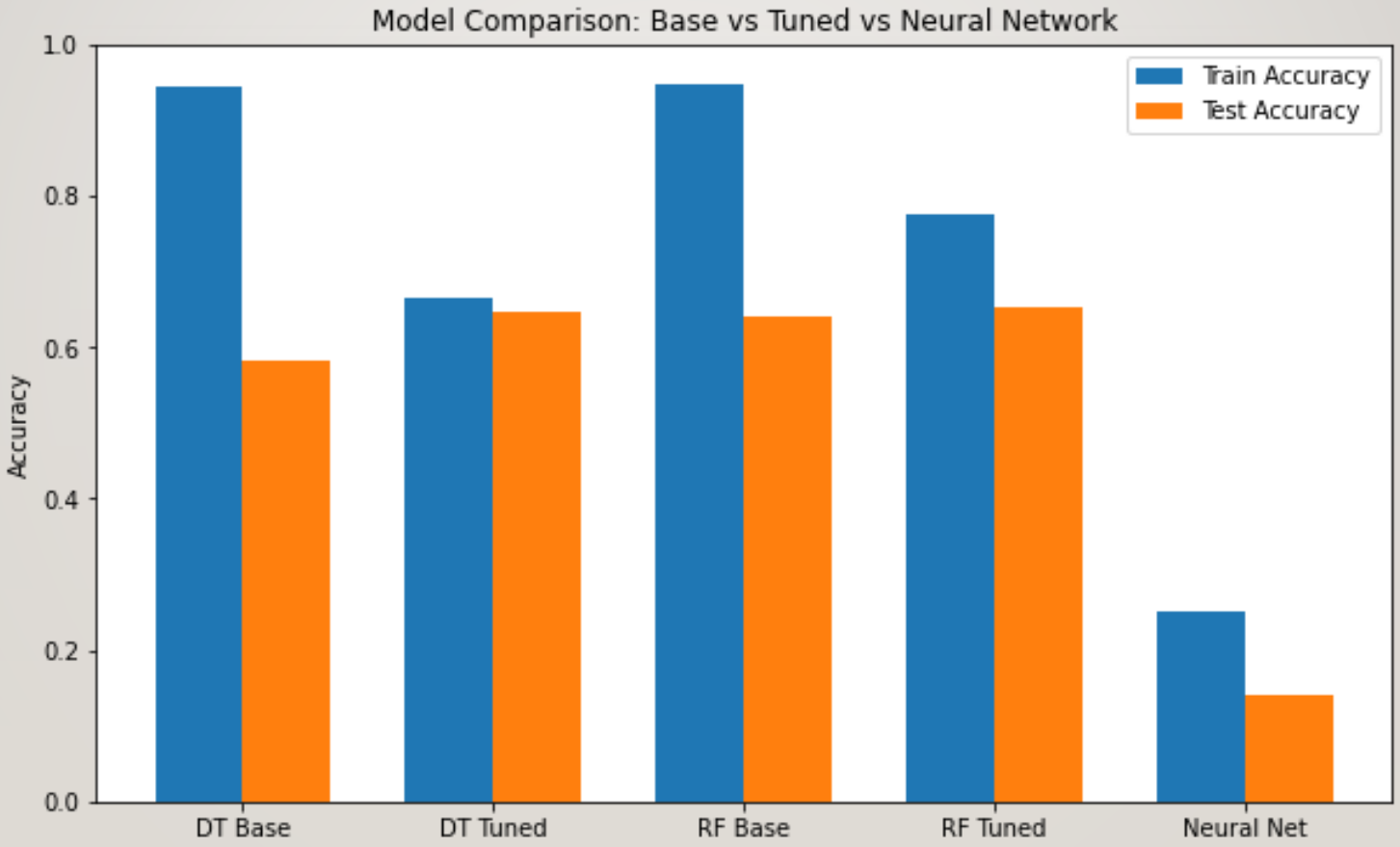
DETERMINING THE RELATION BETWEEN PHYSICAL ATTRIBUTES AND CROPS GROWN.

- Most crops require cultivated land with no erosion, salt, crust, or bare soil
  - Cereals, Vegetables, Rubber → Flatland crops (flat terrain, fine gravel)
  - Fodder plants → Flatland + flood-prone areas (high water association)
  - Fruits, Fibre, Oil-crops, Pulses → Gentle slope crops (tree-linked systems)
  - Roots & Tubers → Irregular crops (sloped land, irregular field patterns)
  - Semi-luxury & Nursery crops → Special cases (tree/shrub systems or coarse, sloping, fragmented land)

DETERMINING THE RELATION BETWEEN FERTILIZER OPTION AND CROPS GROWN.

- **Cereals:** Crop residue + inorganic fertilizer.
- **Fibre crops**: Mainly inorganic fertilizer.
- **Fodder plants**: Crop residue or inorganic fertilizer.
- **Fruits:** Crop residue + animal manure (secondary).
- **Oil-crops & Roots/Tubers**: Mainly crop residue.
- **Vegetables & Semi-luxury crops:** Mixed use of inorganic fertilizer, manure, and crop residue.

MODELLING – PHYSICAL ATTRIBUTES



MODELLING – SOIL IMAGES CLASSIFICATION

- We used a Convolutional Neural Network with:
  - Accuracy – 91%
  - Validation Accuracy – 81%.

DASHBOARD TUTORIAL

RECOMMENDATIONS

- Despite the general recommendations, use AI models for crop prediction.
  - Crop choice depends on multiple soil factors, so machine learning models should guide decisions.
- Promote crop residue as the main fertilizer.
  - Most crops perform best with crop residue, often combined with inorganic fertilizer.

LIMITATIONS OF THE DATASET

- Small image dataset (only 87 usable Kenyan soil images).
  - Limited data for soil image classification + reduced diversity but CNN performed well.
- Severe class imbalance across crop types affected modelling for physical attributes.
- Vertical images contained noise (tools, holes, hands) limiting use.
- Fertilizer recommendations are made based on usage, not yield or soil health

SUGGESTIONS FOR FURTHER STUDIES

- Expand dataset with more soil images across African countries.
    - Include different climates, seasons, and farming conditions.
    - Incorporate crop yield and crop health data.

Thank you